ACDM flavored fried chicken version of Galaxy Dnynamics

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Abstract

"Fried chicken" is a nickname and carries no physics. This paper tunes the knobs of ΛCDM galaxy modelling, supernova feedback first, until halos reproduce the law $g_N = g^2/(g + a_0)$, with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}} = 9.36 \times 10^{-11}\,{\rm m\,s^{-2}}$ and $\kappa = \frac{1}{2}$ fitted, on the measured baryons of 171 SPARC galaxies. It is easy to get close. Twelve feedback knobs, as one global recipe and with no knob set by a galaxy's own curve, land the fitted acceleration within 0.3 % of a_0 and sit 0.0415 dex from the law, 0.0445 on galaxies held out of the tuning. Against the observed rotation curves that recipe scores 0.1953 dex, and the law 0.1951. Adding two concentration knobs gives 0.0275 dex and 0.1933. Six knobs per galaxy, inside 2σ priors, reach 0.0152 dex and pass the 0.02 dex target frozen in advance, though at least half the galaxies then sit at the low bound on concentration. The law's tie to dark energy did not come from the knobs. It took five terms written in by hand, at extreme values, and κ still narrowly missed. A halo can also be written to equal the law exactly, including the tie, by defining it from the formula. That passes every target by construction, and it needs a coupling to dark energy that ΛCDM does not contain. On KiDS-1000 lensing, the law, the recipes and that halo all fail. Version 1.2 tries for a complete ΛCDM recipe, using the law as its basis, against a frozen scoreboard. One global recipe with 26 knobs and hand terms is not complete. Every line can be passed somewhere only by a patchwork of about 1066 parameters, which is not one theory. It also fits X-COP and KiDS better than the plain law does. These recipes were tuned to the law, so a match means freedom, not explanation. This is not evidence for the law or against ΛCDM, and every knob is disclosed.

1 What is being imitated

The target is the law's curve on each galaxy's own baryons, not the observed rotation curve. The law is not new in form: the scale goes back to Milgrom [1, 2], and this interpolating form has appeared before [3]. Only the coefficient looks new, and it is fitted. The law carries its own failures, including IC 1613 and X-COP, listed in PAPER67 (DOI 10.5281/zenodo.23883741). A perfect imitation inherits them.

SPARC [4] supplies the baryons, with $\Upsilon_{\rm disk} = 0.5$, $\Upsilon_{\rm bul} = 0.7$ and no per-galaxy adjustment, giving 3373 points. Four of the 175 galaxies have too few usable points. Against the observed curves on these points, the law's rms is 0.1951 dex.

2 The recipe: twelve supernova knobs

Galaxy simulations are run with subgrid feedback models that carry many tunable parameters. We can't run hydrodynamic simulations here, so R11 is an analytic stand-in, and the mapping from knob to halo is ours, not any simulation's parameter file.

- **Stellar mass outcome, 4 knobs.** The stellar-to-halo relation [9]: normalisation, $\log M_1$, β, γ.
- **Feedback energy, 4 knobs.** $e = \epsilon\,(M_*/0.01M_{200})^{a_1}\,(\Sigma_b/100)^{a_2}\,(f_{\rm gas}/0.5)^{a_4}$, and the core exponent is $n = \tanh e$ [8].
- **Core size, 2 knobs.** $\eta = \eta_0(M_*/10^9)^{a_3}$, core radius $\eta\,R_{1/2}$.
- **Contraction or expansion, 2 knobs.** $f = f_0 + f_1 \log_{10}(M_b/10^{9.5})$ [7].

Halos are Einasto with $\alpha = 0.16$ and $c(M)$ from [10]. The tuned values are log norm -0.319, $\log M_1$ shift $+0.705$, β shift -0.884, γ shift -0.600, $\log\epsilon = -0.639$, $a_1 = -0.760$, $a_2 = +0.091$, $a_4 = -0.097$, $\log\eta_0 = +1.000$, $a_3 = +0.466$, $f_0 = +1.000$, $f_1 = -0.335$. Three sit at the bounds of the ranges we chose: γ, η_0 and f_0. Halo masses stay near abundance matching, with $\log M_{200}$ shifted by -0.23, $+0.11$, $+0.30$ at the 16th, 50th and 84th percentiles.

The recipe gives $a = 9.331 \times 10^{-11}$. Tuned on half the galaxies, it scores 0.0405 dex there and 0.0445 on the other half, so it transfers. With $c(M)$ norm and slope also free (R11b, 14 knobs), it reaches 0.0275, and 0.0326 on unseen galaxies, with M_{200} up by a median of 0.39 dex and both c knobs at their bounds.

3 Targets, frozen before any number

E1: pooled rms to the law at most 0.02 dex. E2: 95th percentile of per-galaxy rms at most 0.04 dex. E3: no half-dex $g_{\rm bar}$ bin off by more than 0.02 dex. E4: $d\ln a/d\ln\rho_{\rm DE} = 0.50 \pm 0.05$ at fixed ω_m, ω_b, A_s. E5: $a(z)/a(0)$ inside the law's band at $z = 1$, 1.5 and 2.33. Most recipes here were added after first numbers, at the owner's request, and are marked Amendments A1 to A3 in `FROZEN_RECIPE.md`. The targets never moved. The knob ranges and priors are our judgement.

4 The menu

Recipe	Parameters	rms	p95	worst bin	a (10^{-11})
R1 NFW, abundance matching, no tuning	0	0.1419	0.2877	0.1613	15.76
R1 tuned, with contraction	4	0.0715	0.1486	0.0441	9.12
R2 tuned, with feedback core	6	0.0721	0.1357	0.0283	8.65
R3 thermal halo, two global laws	2	0.2245	0.3187	0.2261	6.06
R4 thermal halo, per galaxy	342	0.0609	0.1119	0.0260	8.97
R7 surface-density coupling	9	0.0408	0.0777	0.0413	9.21
R11 twelve feedback knobs	12	0.0415	0.1006	0.0221	9.33
R11b feedback plus c norm, slope	14	0.0275	0.0624	0.0196	9.31
R10 six knobs as global functions	42	0.0213	0.0441	0.0216	9.30
R9 per galaxy, 1σ priors	1026	0.0376	0.0711	0.0164	9.63
R9 per galaxy, 2σ priors	1026	**0.0152**	**0.0283**	**0.0040**	9.43
R8p per galaxy, referee priors	684	0.0396	0.0665	0.0148	9.72
R8 per galaxy, wide	684	**0.0056**	**0.0130**	**0.0031**	9.36
R6 one empirical function	1	**0.0075**	**0.0118**	**0.0180**	9.37
R5 exact shell masses	3373	**0.0000**	**0.0000**	**0.0000**	9.36
R12 halo defined from the formula	1	**0.0000**	**0.0000**	**0.0000**	9.36

Table 1: Distance to the law in dex. Bold passes E1 to E3. Thresholds are 0.02, 0.04 and 0.02.

The level comes cheaply. Four knobs bring the fitted acceleration from 15.76 to 9.12×10^{-11}, within 3 % of a_0, with the concentration knob at its bound. Getting the right number from tuned halos is easy, so the number alone proves little.

The shape does not. Halos set by mass alone can't follow galaxies of the same mass and different surface density. Coupling c and M_{200} to M_b/R_d^2 helps, to 0.0408 dex, with the abundance-matching normalisation at its bound. It still misses all three targets. Measured against the observed curves, R7 has rms 0.1964 dex, R11 0.1953, and R11b 0.1933, against the law's 0.1951. None was tuned to those curves. With these fixed mass-to-light ratios, this statistic barely tells them apart, so a good fit to rotation curves alone cannot pick the law over a tuned halo.

The thermal halo misses. An isothermal halo in the baryon potential looked like the natural imitator [13]. With two global laws it fails at 0.2245 dex, from either starting point, and with 342 free parameters it still reaches only 0.0609. Its per-galaxy fits do land near $\sigma^4 = GM_b a_0/4$, median ratio 1.017, with wide scatter (0.568 to 3.174).

5 The exact dish, and its bill

R9 is the closest match on ΛCDM parts inside stated priors: per galaxy M_{200}, c, Einasto α, contraction fraction, core exponent and core radius. With 2σ priors (0.22 dex in c, 0.6 dex in M_{200}) it passes E1 to E3 at 0.0152 dex. The bill: median $\log c$ shift -0.22, at the bound, and median α 0.1, also at the bound. Scatter can't push half the galaxies to the same edge, so this is really a population shift in c. With 1σ priors it fails, at 0.0376. NFW [6], Blumenthal contraction [7], a coreNFW core [8]. Each galaxy gets its own M_{200}, c, contraction fraction and core exponent. R8 passes E1 to E3. The bill, relative to abundance matching [9] and $c(M)$ [10]: $\log M_{200}$ shifts by median $+0.29$ and $+1.2$ at the 84th percentile, and $\log c$ by median -0.42, reaching -0.81 at the 16th. Contraction is near full (median 0.9). Under priors of 0.11 dex in c and 0.3 dex in M_{200}, R8p is 0.0396 and fails E1 and E2.

R10 writes those six knobs as quadratic functions of M_b, surface density and gas fraction: 42 coefficients, one recipe. It stops just short at 0.0213 dex, and 0.0270 on unseen galaxies, but with median $\log M_{200}$ shift $+0.46$, $\log c$ shift -0.74, and 0 of 171 galaxies inside 1σ of both relations.

R6 is one line, the published empirical form [5], with $g_\dagger = 9.585 \times 10^{-11}$, or 1.024 a_0. It passes E1 to E3 with one number, but writes the law's content as a halo response with no process behind it.

R5 is the law renamed: $M_{\rm DM}(r) = r^2(g_{\rm law} - g_{\rm bar})/G$. We checked whether it could exist at all, as a spherical halo with baryons smoothed by 0.1 dex. Density is positive in 171 of 171, the isotropic Jeans dispersion is positive in 171 of 171, and the Eddington distribution function is non-negative to 1 % of its peak in 171 of 171. The same Eddington code returns $+0.0000$ for Hernquist and -0.2555 for a cored tracer in a point-mass potential, so it can fail. Stability is therefore no argument against the imitation.

Its origin is the weak point. Undoing Blumenthal contraction gives a monotonic initial profile in 158 of 171 galaxies. An NFW fit to that initial profile has median rms 0.057 dex and 0.261 at the 84th percentile, with median c at 0.32 of the $c(M)$ value. Against abundance matching, the implied $\log M_{200}$ has median $+0.16$, but -0.76 at $\log M_b$ 11 to 12.

6 The dark energy tie

This is the part the knobs don't supply. On R10 with no hand term, the slope with $\rho_{\rm DE}$ is $+0.283$, κ runs 0.459, 0.614, 0.735 across $(\Omega_\Lambda, H_0) = (0.70, 67.4)$, $(0.685, 73)$, $(0.65, 73)$, and $a(z)/a(0)$ is

1.58, 2.12, 3.44 at $z = 1$, 1.5, 2.33. Five terms written in by hand change that. $c \propto \rho_{\rm DE}^{3.00} H_0^{-1.61}$ and $M_{200} \propto H_0^{-12}$ give slope +0.496, which passes, and κ of 0.485, 0.506, 0.525, a narrow miss of the 5 % target. Two of those exponents sit at the bounds we allowed, and no process produces them. $c \propto (1+z)^{-0.26-0.89z}$ gives 0.93, 0.91, 0.92, inside the band, with baryons held fixed in redshift, which no real galaxy does. So the tie can be imitated by hand, almost, and then it is the law's physics written into the halos. R5 and R6 pass E4 and E5 only if a_0 is written into them directly. The thermal halo R3, with the same scale written in, gives +0.810 and fails.

7 The halo exactly like the formula

R12 is the reverse-engineered dish: one function for every galaxy, added after the owner asked for it and frozen as Amendment A6.

$$M_{\rm DM}(r) = \frac{r^2}{G}\left[g_{\rm law}(g_{\rm bar}; a) - g_{\rm bar}\right], \qquad a = \kappa c\sqrt{G\rho_{\rm DE}}, \quad \kappa = \tfrac{1}{2} \text{ by hand.} \tag{1}$$

In ΛCDM's language this is a halo of collisionless dark matter particles, with the particle left unspecified. Only the mass profile enters, so the model says nothing about the particle's mass or identity. R5's Eddington check is what allows that reading: a spherical, isotropic, collisionless equilibrium exists for this profile inside the measured radii. Particles appear here only as the rival's language. The framework this paper imitates has none.

It passes everything by construction. Its rms to the law is 0.0000 dex, its rms to the observed curves is 0.1951, κ is 0.500 at all three cosmologies, the slope with $\rho_{\rm DE}$ is +0.500, and $a(z)/a(0)$ is 1.000 for constant $\rho_{\rm DE}$, inside the band. None of that is evidence. The formula went in, so the formula comes out.

The bill has three items. First, a particle halo has no term through which it could know $\rho_{\rm DE}$, so the tie needs a new coupling that is not one of ΛCDM's knobs. Second, no formation path is supplied, and the origin problem from R5 stands. Third, implied $\log M_{200}$ sits a median +0.11 dex from abundance matching (16th to 84th percentile −0.23 to +0.27), but −0.80 at $\log M_b$ 11 to 13. Baryon fractions are not a problem: M_b/M_{200} stays below the cosmic 0.156 in all 171.

8 Lensing

Added after version 1.0 as Amendments A4 to A6, each frozen before its numbers. Data are the KiDS-1000 profiles of isolated lenses [11]: four stellar mass bins, 60 points from 35 kpc to 2.6 Mpc, full covariance, and nothing refitted. The gate is $p \geq 0.01$. Recipes carry a Tinker bias [12] two-halo term and stop at R_{200}. The plain law has neither. The recipes, tuned only on SPARC, do not lose

Model	χ^2, 60 points	χ^2, $R \leq 1$ Mpc, 44 points	amplitude A
Plain law, and R12 untruncated	276.1	211.4	1.367
R12 stopped at R_{200}, two-halo	457.7	290.4	1.422
R11 twelve knobs	277.9	146.5	1.179
R11b fourteen knobs	242.1	120.9	1.140
Base ΛCDM control, untuned	563.1	444.2	0.809

Table 2: All rows fail the gate. A is one fitted scale factor, a diagnostic only; 1 is no offset.

to the law here, and a halo made more realistic by a cutoff does worse. A post-result diagnostic, with a free mass shift in each bin and so not a grade, gives the law 167.7 on 56 degrees of freedom

at +0.2 to +0.3 dex, R11 228.2 and R11b 206.0. That law number is close to an earlier, separately coded result on the same file, 163.1, in the repository's lane CFG801. All rows may share a problem with lens masses or the covariance. Satellites and the isolation cut are not modelled, recipes are built at $z = 0$, and these lenses lie mostly outside the tuned range.

Two more checks. The R11 and R11b fits are converged at 0.0415 and 0.0275 from their recorded knobs, which is a local statement. Independent halo scatter of 0.11 dex in c and 0.15 dex in M_{200} raises their rms to observed curves only to 0.2032 and 0.1999, passing a limit of 0.205 set in advance, though noise dominates that statistic.

9 The high priest's complete recipe

R13 is one global recipe tuned jointly to the law on SPARC, to KiDS and to X-COP hydrostatic masses, frozen as Amendment A10: R11b's 14 knobs, an AGN core cutoff, lens mass offset, two-halo multiplier, satellite fraction and host mass, hydrostatic bias, and cluster baryon fraction, 21 in all, plus 5 dark energy and redshift hand terms. It was tuned on half of each dataset and graded on the rest, then tuned on all. The first run had an X-COP radius error of a factor 1000 and two smaller errors. They were fixed under the same gates and the wrong output is kept. Hand terms remain

Line	R13	Result
L1 distance to the law, limits 0.02, 0.04, 0.02	0.0290, 0.0640, 0.0187	fail, E3 passes
L2 rms to observed SPARC, limit 0.205	0.1943	pass
L3 KiDS, 60 points, $p \geq 0.01$	145.2	fail
L4 X-COP mean, rms, limits 0.05, 0.12	−0.001, 0.075	pass
L5 slope with $\rho_{\rm DE}$, κ, with hand terms	+0.500; 0.483, 0.49, 0.51	pass
L6 $a(z)/a(0)$, with hand terms	0.929, 0.913, 0.92	pass
L7 Cassini, smooth halo	$2.0 \times 10^{-9}\, a_0$	pass
L8 unseen: SPARC, KiDS χ^2, X-COP	0.0281, 82.1 on 30, 0.048	fail on KiDS
L9 knobs at bounds, limit 3; median $\log M_{200}$ shift, limit 0.3	7; 0.32	fail

Table 3: R13's frozen scoreboard. Verdict: not complete.

extreme: $c \propto \rho_{\rm DE}^{3.00} H_0^{-4.86}$, $M_{200} \propto H_0^{-9.54}$. Without them the slope is +0.245. The X-COP pass rests on a cluster baryon fraction at its lower bound of 0.08, while the sample's measured value in one system is 0.138, and hydrostatic bias at 0. That is a poor physical reading, so L4's pass is weak.

R14 is the escalation, frozen as Amendment A11 after R13's verdict. SPARC uses R9, 1026 per-galaxy parameters. Lensing adds 14 per-bin parameters on R13's halo. It reaches $\chi^2 = 69.6$ on 60 points, p 0.014 at 46 degrees of freedom, and on unseen radii 44.7 on 28, p 0.024. Four of the 14 are at bounds, including truncation at $3\,R_{200}$ and the host mass. Counting R13's 21 knobs and 5 hand terms gives about 1066 parameters. It is a patchwork: nothing connects R9's per-galaxy values to the lens side, and the hand terms were not rechecked on R9's halos.

So the honest reading has two parts. Getting close is easy. Getting everything at once, as one theory, was not achieved here. For comparison, the plain law on the same X-COP points has mean −0.467 and rms 0.476 dex, and on KiDS 276.1 against R13's 145.2. On these two datasets the tuned recipe fits the data better than the law. Not graded: CMB and growth on the recipe, cluster keys, dwarfs, IC 1613, DiskMass, the Milky Way, the Bullet Cluster, wide binaries, and any formation path.

10 If the halo were a particle

This is an effective reading in the rival's language. The author does not hold that the dark sector is a particle, and these results don't show that one exists. A static scalar with $y^2/(a_0 - y) = g_{\rm bar}$ reproduces the law to 3.9×10^{-13} on all 3373 points, in spherical symmetry, so $y < a_0$ always. If instead a real particle halo supplies all of $g_{\rm law} - g_{\rm bar}$, no force acts on baryons and double counting goes away. That halo can't bind to the Sun beyond 0.076 AU, so Cassini passes, and wide binaries should be Newtonian, unlike a field reading of the law.

A boson that condenses in all of SPARC and not in clusters at 100 kpc lands between 9.2 and 13.7 eV on our thresholds. But $\log P$ at fixed $\log \rho$ scatters by 0.521 dex across galaxies, against a limit of 0.1, so no single equation of state holds these halos up, which undercuts a simple condensate. A collisionless route from an NFW start passes the mixing theorem in 139 of 171 galaxies, all failures below $\log M_b = 10$; a frozen control there failed because our expectation was wrong, and is disclosed. The halos need a median 0.516 of all supernova energy at full coupling. Tracking baryons by scattering would need σ/m of 0.63 to 40 $\rm cm^2\,g^{-1}$. A fermion needs a mass above about 56 eV for a typical galaxy and 245 eV for the most demanding. No part of this derives $\rho_{\rm DE}$ in the halo.

11 Reading the menu

The point of the exercise is how little it takes. A dozen feedback knobs get the level, nearly the shape, and a tie with the law on observed curves. So a future ΛCDM paper showing a tight relation at this scale has not, on that alone, derived anything. A paper claiming to derive this law from ΛCDM should show these five things: the prior on M_{200} and c per galaxy, the fit against surface-density diversity and not only mass, the slope with $\rho_{\rm DE}$ with baryons evolved, κ across cosmologies, and a formation path to the required halo. Failing them doesn't refute ΛCDM, since real simulations could do things this model can't. Passing them would be a real result, and the law would then have to be read as an output of the halo side.

12 Limits

This is a semi-analytic model, not a simulation. Baryons are spherically approximated in the halo response. Mass-to-light ratios are fixed. Priors and knob ranges are our judgement. The feedback mapping is ours and has not been tested against any simulation, so whether real subgrid physics can reach these knob values is open. Masses M_* use one Υ and one SHMR. Among clusters only X-COP is graded. A coding error in the cosmology rows was found and fixed during this work; the wrong output is kept in the repository. Three coding errors in the lensing two-halo code were fixed before its first full table. All numbers are AI-run and have not been reproduced independently.

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